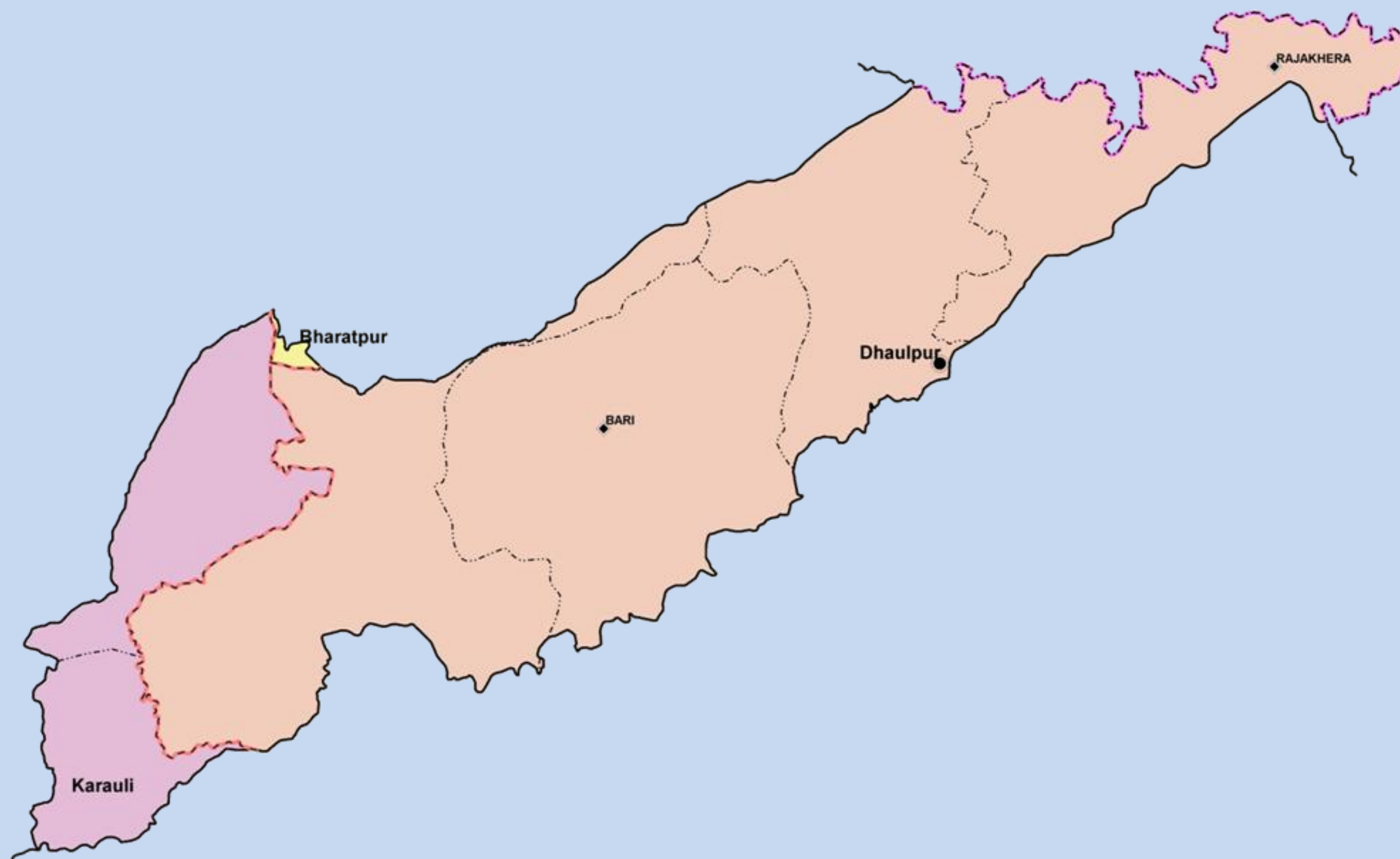


Hydrogeological Atlas of Rajasthan

Parbati River Basin



2013

Message



Shri Ashok Gehlot,
Hon'ble Chief Minister, Rajasthan

Ground water is an inseparable aspect in the lives of the people of Rajasthan. There are no major or perennial rivers flowing through the State. Moreover, most part of the state receives very less rainfall. Our dependence on ground water is extremely critical not only as drinking water for human beings and cattle but also as major source of irrigation.

The Government of Rajasthan is keen on sustainable utilization of the ground water. The State water policy, adopted in 2010 emphasizes on community participation in ground water management and use of information technology for widespread dissemination of data for the benefit of all sections of the society.

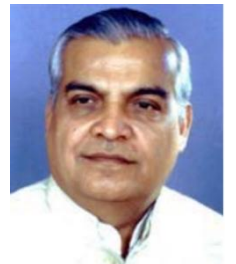
I am pleased to note that the Ground Water Department of the State has carried a detailed assessment of quality, quantity and distribution of ground water resources of the State through mapping of aquifers and using most modern tools and techniques.

It is also heartening to note that the outcome of the project is being published in the form of '*Hydrogeological Atlas of Rajasthan*'. It will incorporate the scientific findings and reliable information in various types of readily understandable GIS maps based on ground water potential viz. zone maps, depth to water level maps, ground water categorization maps, chemical quality maps, aquifer maps etc. These would be certainly found useful by experts and user agencies. The same would also help in raising the awareness levels of the various stakeholders in utilizing the precious ground water resources of the State.

I take this opportunity to congratulate the Ground Water Department officials for bringing this useful publication and wish them success in all such endeavor.

(Ashok Gehlot)

Message



Dr. Jitendra Singh,
Hon'ble Minister for Energy,
Non-Conventional Energy Resources,
PHED, GWD, Information & Public Relations

Ground water is a renewable resource but its long term sustainability depends on the balance of recharge and withdrawal. Ground water is added every year through rainfall and other sources. In Rajasthan, volume of water extracted for different purposes far exceeds its replenishment which has led to depletion of ground water in large parts of state.

While the surface water is visible and largely measurable, the ground water is hidden underneath. To meet the ever increasing demand of water, the state has been promoting conjunctive use of water. In order to have a state level estimation of water resources, the estimation of ground water quality and quantity is a key input. The studies being very scientific in nature it remains a challenge how to communicate the message to all stakeholders.

The publication of '*Hydrogeological Atlas of Rajasthan*' is a major step forward and will help in visualization of ground water related information in very simplified graphic and tabular forms. The effort that has gone into creation of the atlas from basic information in archives is very much appreciated.

(Jitendra Singh)

Message



Dr. Purushottam Agarwal

Principal Secretary to Government,

Public Health Engineering and Ground Water Department, Rajasthan

Ground water has been the principal source over years to meet the needs of human and cattle population of the state. Increasingly modern techniques are used for extraction of large volume of ground water which has exceeded recharge in most parts of the State. As a result, water table in most of the areas has gone down significantly. People are digging deeper and deeper to reach the ground water to meet their increasing needs. Maintaining a healthy practice of balance between recharge and withdrawal is critical for its sustainability.

The state had initiated various programs under which a systematic assessment of water resources is being carried out and all future water management plans are chalked out on the basis of sound database and historic data analysis. Increasing population, industrialization, agricultural need, power generation etc. are bound to pose challenges before us in coming years and to face them we have documented 'Water Resource Vision 2045' which highlights the short term (upto 2015) and long term (upto 2045) thrust areas and action plan which are pre-requisites for successful implementation of the State Water Policy. We are committed to achieving the objective of optimum use of every drop of scarce and precious utilizable water resource. In this regard, the Ground Water Department has also carried out a systematic assessment of ground water resources and three dimensional mapping of aquifers that gives a new insight.

I congratulate the Ground Water Department on publication of '*Hydrogeological Atlas of Rajasthan*' which emphasizes on various aspects of ground water which I feel will help us in spreading the information on current scenario and consequently help in managing it better.

(Purushottam Agarwal)

Foreword



Arno Schaefer

Minister Counselor, Head of Operations

The Government of Rajasthan has entered into an agreement with the European Union through the State Partnership Programme for implementing state-wide water sector reform leading to sustainable integrated water resources management in Rajasthan, and supporting the Panchayat Raj institutions in executing their responsibilities in equitable access to safe, adequate, affordable and sustainable water supply as well as conservation, stabilization and replenishment of surface and ground water.

Ground water is the primary source for drinking, domestic, agriculture and industries and other purposes in Rajasthan. The stage of ground water development is close to 135% i.e., if the net annual ground availability in Rajasthan is about 10.7 billion cubic meter, the annual draft is 14.5 billion cubic meter. As a result of this, the water levels have been depleting alarmingly. The ground water in 166 out of 249 blocks of the state fall under the category of over-exploited. This indiscriminate and sometime exploitative use of ground water has led to questions regarding its sustainability.

This publication '*Hydrogeological Atlas of Rajasthan*' is the culmination of a long term effort of collation, integration, interpretation and analysis of vast archive of historic data obtained by Rajasthan Ground Water Department. The atlas is in three parts, (1) Statewide Atlas, which shows the state of ground water, water quality, current stage of exploitation, aquifer mapping and basic information on administrative setup, climate, geology and geomorphology; (2) A detailed series, wherein compilation of information and district level thematic maps and all 33 districts have been separately presented and (3) Compilation of information at the River Basin level. In addition to the thematic maps, a large number of cross sections and the 3 dimensional features of aquifers have also been presented.

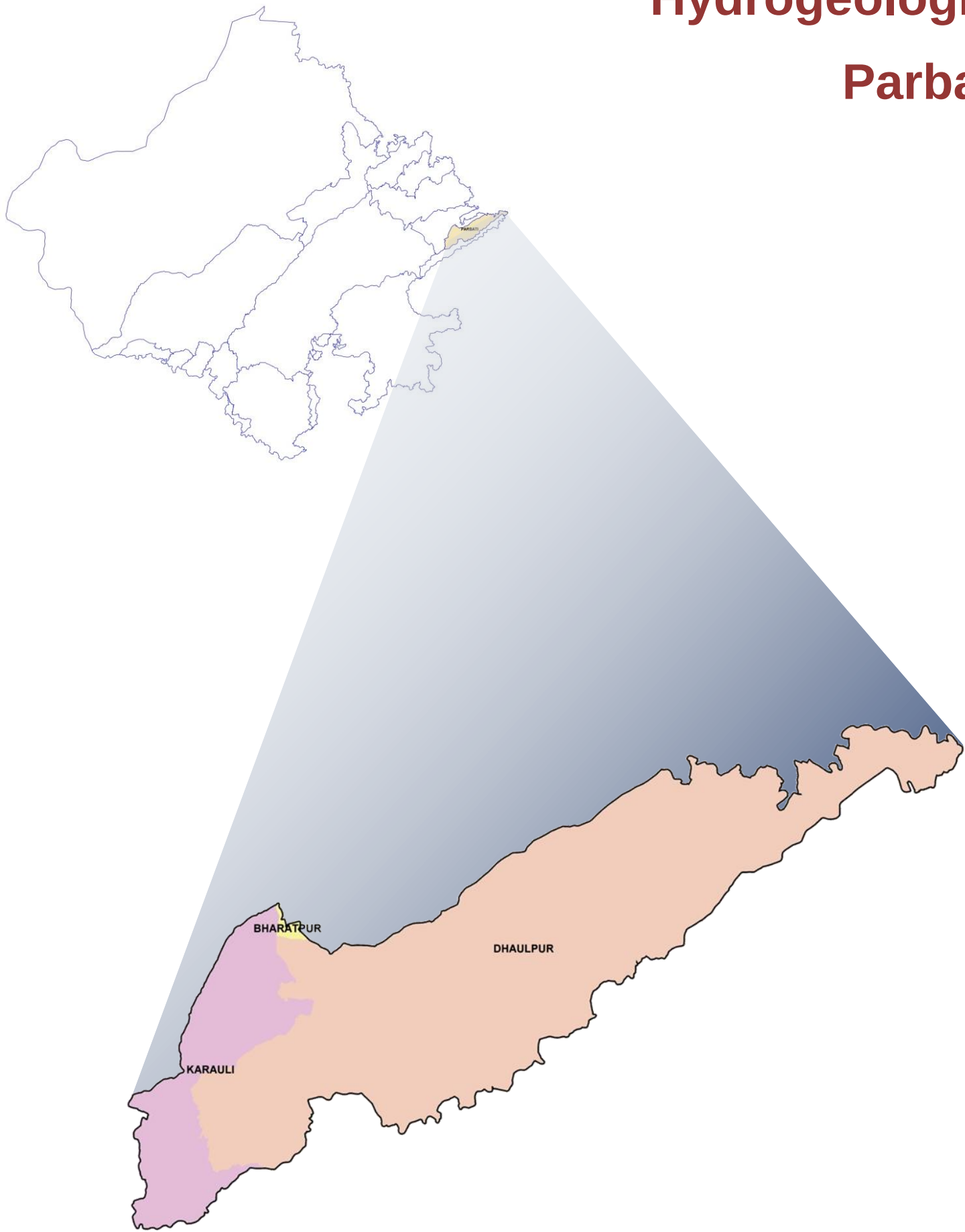
I am convinced that the Atlas will contribute to better understand ground water and help managerial and future planning. The current Geographic Information System (GIS) based study is a pioneering step in India and will pave the way for similar approach to be adopted by other Indian States.

I appreciate very much the efforts made by the Ground Water Department, Government of Rajasthan, The European Union Technical Assistance Team and M/s Rolta India Limited for undertaking this study and compiling this Atlas under the India-EU State partnership Programme.

(Arno Schaefer)

Hydrogeological Atlas of Rajasthan

Parbati River Basin



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ADMINISTRATIVE SETUP

PARBATI RIVER BASIN

Location:

The Parbati River Basin is located in the eastern part of Rajasthan. It stretches between 26° 20' 29.76" to 26° 58' 05.87" North latitude and 77° 08' 33.59" to 78° 16' 27.33" East longitudes. The basin extends in NE-SW direction and is bounded by the Gambhir River Basins in the northwest side and the Chambal River Basin in the southeast. The basin extends over parts of Bharatpur, Dhaulpur and Karauli Districts. The total catchment area of the Basin is 2,266.2 km².

River Parbati originates in the hills in Karauli district close to Sawai Madhopur District border. It flows in a generally northeast direction for about 123 km before joining Gambhir River near Kharagpur village in Dhaulpur District. The main tributaries of the river are Sairni on the left side, and Bamni, Mendka on the right side.

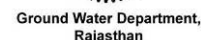
Administrative Set-up:

Administratively, Parbati River Basin extends over parts of Bharatpur, Dhaulpur and Karauli districts encompassing 7 Blocks divided into 584 towns and villages. While the basin has very small contribution from Bharatpur district, the largest part of basin contributor is Dhaulpur district which contributes approximately 83% of its whole area.

S. No.	District Name	Area (sq km)	% of Basin Area	Total Number of Blocks	Total Number of Towns and Villages
1	Bharatpur	7.1	0.3	1	1
2	Dhaulpur	1,873.8	82.7	4	549
3	Karauli	385.3	17.0	2	34
Total		2,266.2	100.0	7	584

Climate:

The basin is located on the eastern fringe of Rajasthan in close proximity to Uttar Pradesh and Madhya Pradesh. It falls within the semi-arid to sub-humid regions but witnesses extreme temperatures during the hottest months which are May and June. Coldest months are December and January where temperatures sometimes reach near-zero and subzero levels. In some years, the winters and summers (recorded -4.3°C in Jan 1990 and 50°C in June 1995 respectively) cross the extremes and record such unusually high and low temperatures. The rain are mostly received during monsoon season i.e., from July through September, and mean annual rainfall over the Parbati Basin was computed as 638 mm.



TOPOGRAPHY

The western part is hilly which is the area where the Parbati river originates, and then the topography slopes towards northeast. The hills mostly belong to the Aravali ridges and have a southwest – northeast trend and the river flowing in the intervening valleys also follows the same trend till it drains out of the basin and enters Uttar Pradesh eventually to join the Yamuna river. The elevation ranges from a minimum of 135m in Dhaulpur district to a maximum of about 379m in Karauli district. The undulating plains between Bari and Rajkhera often show the development of ravines especially when closer to Uttar Pradesh border.

Table: District wise minimum and maximum elevation

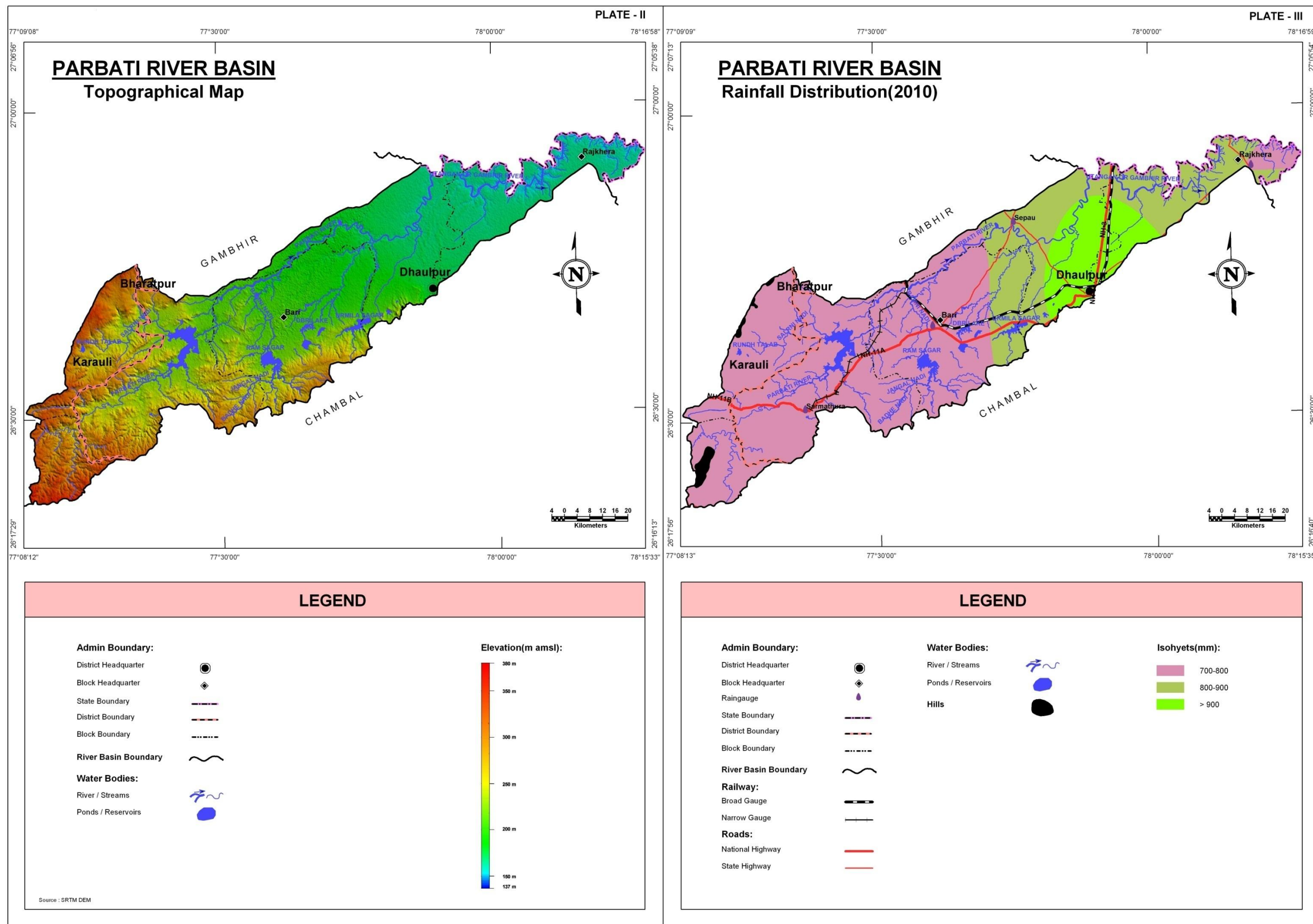
S. No.	District Name	Min Elevation (m amsl)	Max. Elevation (m amsl)
1	Bharatpur	277.0	325.4
2	Dhaulpur	135.4	340.5
3	Karauli	221.1	378.7

RAINFALL

The general distribution of rainfall across the Parbati River Basin can be visualized from isohyets presented in the Plate III where the western part receives lesser rainfall (minimum total annual rainfall of 712mm) which seem to increase gradually towards east where it reaches a high of upto 975mm. The average annual rainfall computed based on available station data is about 829 mm. The rainfall data for available rain gauge stations is presented below.

Table: District wise total annual rainfall (based on year 2010 meteorological station recordings (<http://waterresources.rajasthan.gov.in>))

S. No.	Rain gauge Stations	Total Monsoon Rainfall (mm)	Total Non-Monsoon Rainfall (mm)	Total Annual Rainfall (mm)
1	Bari	600.0	112.0	712.0
2	Dholpur	895.0	80.5	975.5
3	Rajakhera	742.0	56.0	798.0



GEOLOGY

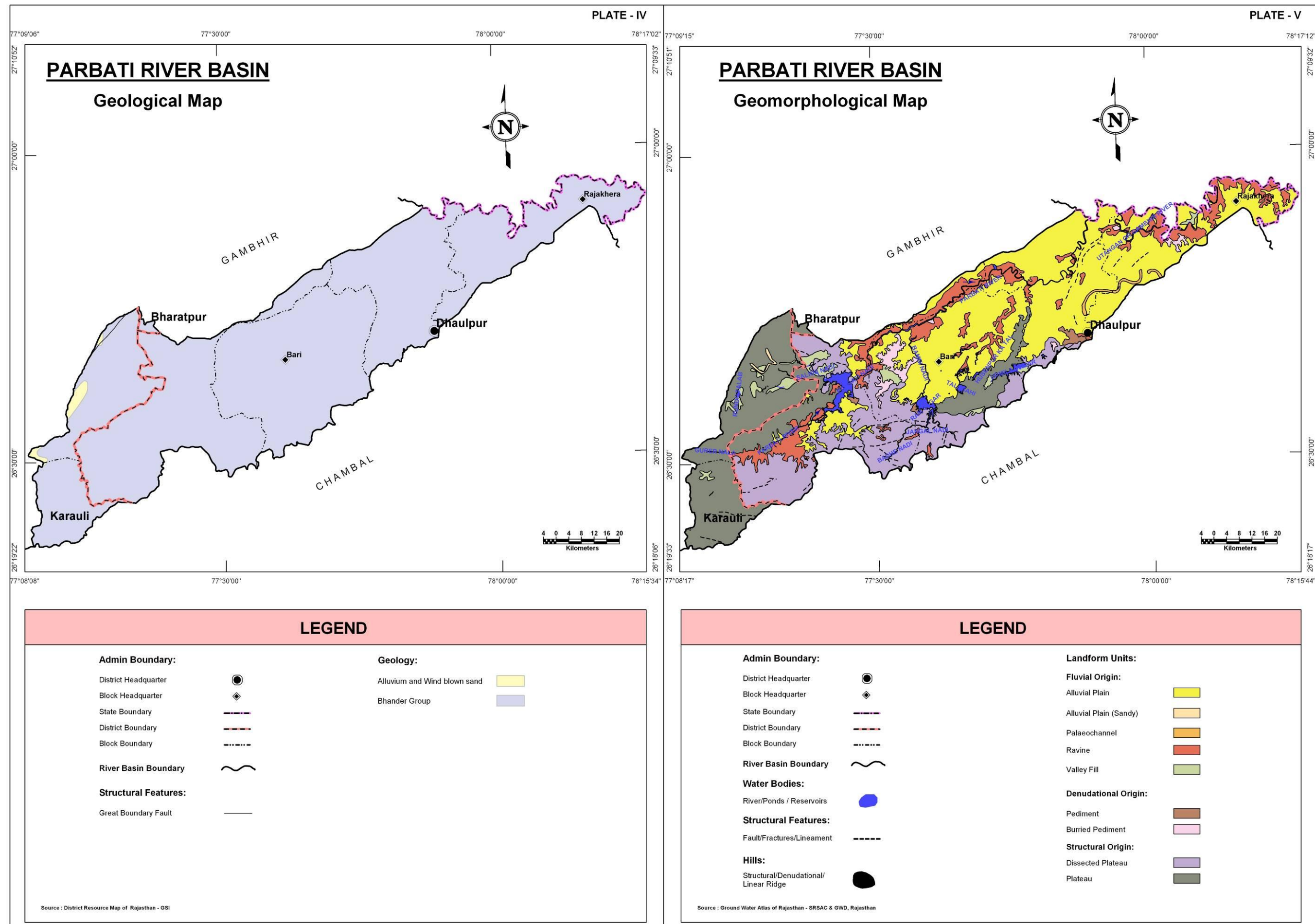
PARBATI RIVER BASIN

The area covered by the basin is very small (<2300 sq km) so geologic variation in the basin is also very limited. The Vindhyan (Sandstones, Shales and Limestones) represent older rocks in the area whereas the older alluvium represents youngest geologic formations in the area consisting of clay, sand and gravel with kankar.

Age	Super-Group	Group/ Formation	Rock Types
Holocene Pleistocene	----	Older Alluvium	Wind-blown sand and fluvial deposits, alluvial clay, silt, with clay, sand and sand and gravel with kankars
-----X-----X-----Unconformity-----X-----X-----			
Upper Proterozoic	Vindhyan	Bhander (Upper)	Sandstone, Shale and Limestone

GEOMORPHOLOGY

Origin	Landform Unit	Description
Denudational	Buried Pediment	Pediment covers essentially with relatively thicker alluvial, colluvial or weathered materials.
	Pediment	Broad gently sloping rock flooring, erosional surface of low relief between hill and plain, comprised of varied lithology, criss-crossed by fractures and faults.
Fluvial	Alluvial Plain	Mainly undulating landscape formed due to fluvial activity, comprising of gravels, sand, silt and clay. Terrain mainly undulating, produced by extensive deposition of alluvium.
	Alluvial Plain (Sandy)	Flat to gentle undulating plain formed due to fluvial activity, mainly consists of gravels, sand, silt and clay with unconsolidated material of varying lithology, predominantly sand along river.
	Paleochannel	Mainly buried on abandoned stream/river courses, comprising of coarse textured material of variable sizes.
	Valley Fill	Formed by fluvial activity, usually at lower topographic locations, comprising of boulders, cobbles, pebbles, gravels, sand, silt and clay. The unit has consolidated sediment deposits.
	Ravine	Small, narrow, deep, depression, smaller than gorges, larger than gulley, usually carved by running water.
Structural	Dissected Plateau	Plateau, criss-crossed by fractures forming deep valleys.
	Plateau	Formed over varying lithology with extensive, flat, landscapes, bordered by escarpment on all sides. Essentially formed horizontally layered rocky marked by extensive flat top and steep slopes. It may be criss crossed by lineament.
Hills	Denudational, Structural Hill, Linear Ridge	Steep sided, relict hills undergone denudation, comprising of varying lithology with joints, fractures and lineaments. Linear to arcuate hills showing definite trend-lines with varying lithology associated with folding, faulting etc. Long narrow low-lying ridge usually barren, having high run off may form over varying lithology with controlled strike.



AQUIFERS

PARBATI RIVER BASIN

Surface cover is mostly the alluvium but interestingly, when it comes to aquifers, it is the Vindhyan Sandstone which occupies about 63% of the aquifer area in the basin followed by Older Alluvium (37%).

Aquifer in Potential Zone	Area (sq km)	% of Basin Area	Description of the unit/Occurrence
Older Alluvium	829.0	36.6	This litho unit comprises of mixture of heterogeneous fine to medium grained sand, silt and kankar.
Vindhyan Sandstone	1,420.3	62.7	Fine to medium grained, red colour and compact and at places.
Non Potential Zone	16.9	0.7	(Hills
Total	2,266.2	100.0	

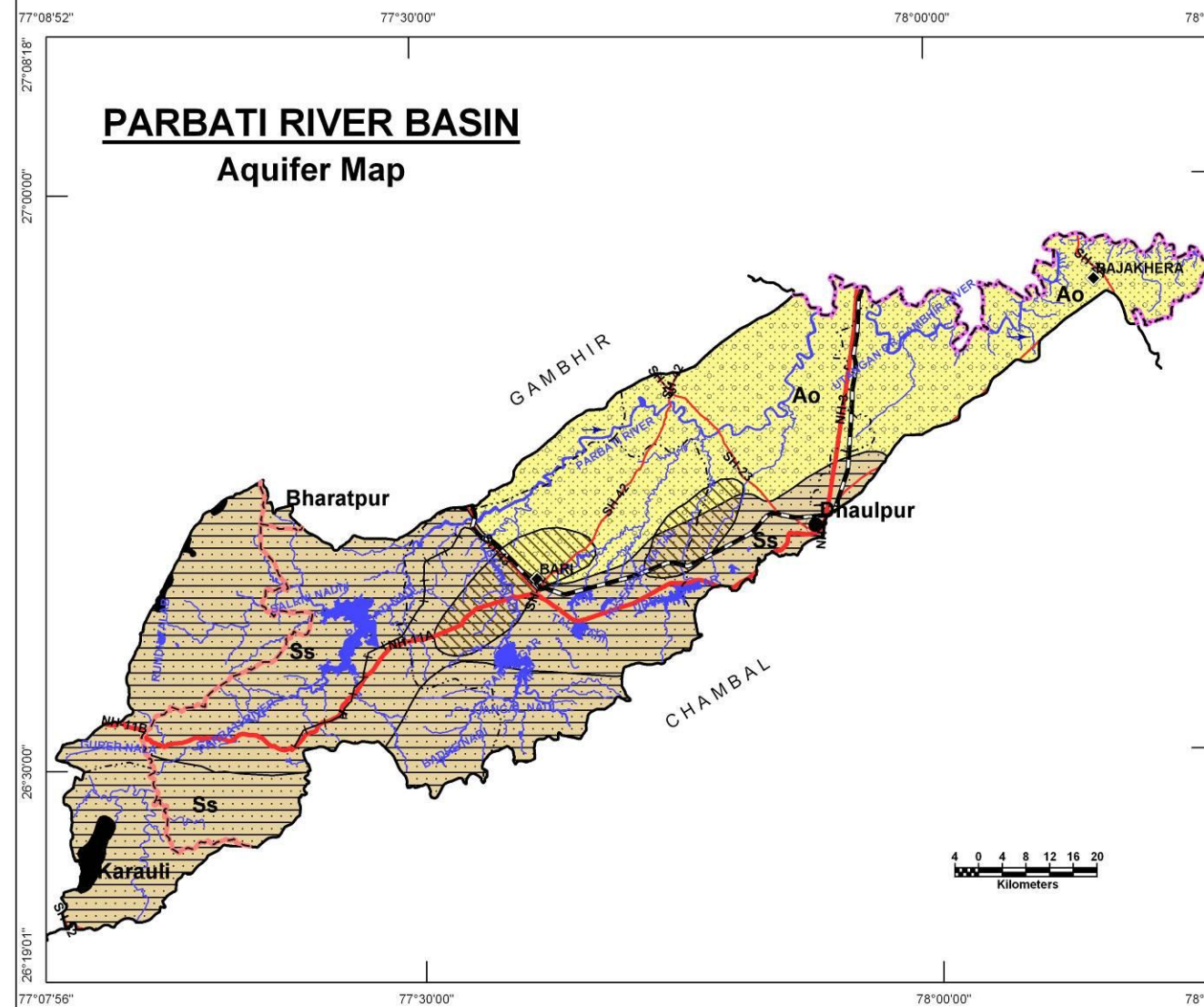
LOCATION OF GROUND WATER MONITORING WELLS

The basin has a well distributed network of large number of ground water monitoring stations (86) in the basin owned by RGWD (74) and CGWB (12); and an additional 38 wells have been recommended to be added to network to effectively monitor ground water quality (34 additional wells) and water level (4 additional wells) in the basin.

District Name	CGWB	RGWD	Total	Recommended additional wells for optimization of monitoring network	
				Water Level	Water Quality
Bharatpur	-	-	-	-	-
Dhaulpur	12	74	86	4	33
Karauli	-	-	-	-	1
Total	12	74	86	4	34



PLATE - VI



78°15'34\"/>

LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- State Boundary
- District Boundary
- Block Boundary
- River Basin Boundary
- Roads:**
 - National Highway
 - State Highway
- Railway:**
 - Broad Gauge
 - Narrow Gauge

Water Bodies:

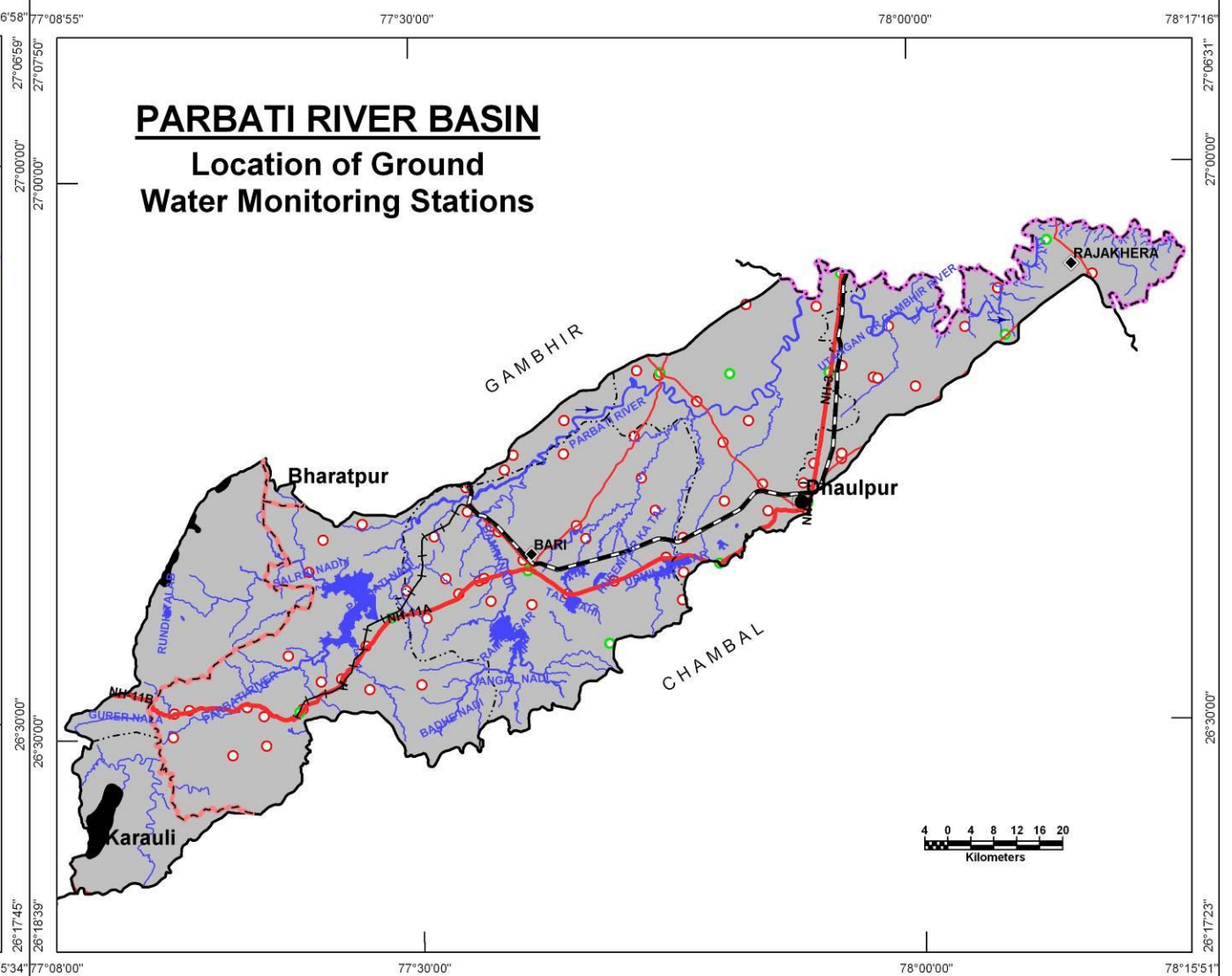
- River / Streams
- Ponds / Reservoirs
- Hills
- Command Area

Aquifer:

- Older Alluvium
- Sandstone

Source: Ground Water Potential Zone Map - GWD, Rajasthan

PLATE - VII



78°15'51\"/>

LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- State Boundary
- District Boundary
- Block Boundary
- River Basin Boundary
- Roads:**
 - National Highway
 - State Highway
- Railway:**
 - Broad Gauge
 - Narrow Gauge

Water Bodies:

- River / Streams
- Ponds / Reservoirs
- Hills

Ground Water Monitoring Stations:

- CGWB
- RGWD

LOCATION OF EXPLORATORY WELLS

In all there are 41 exploratory wells present in the basin drilled in the past by RGWD (32) and CGWB (9) that form the basis for delineation of sub-surface aquifer system.

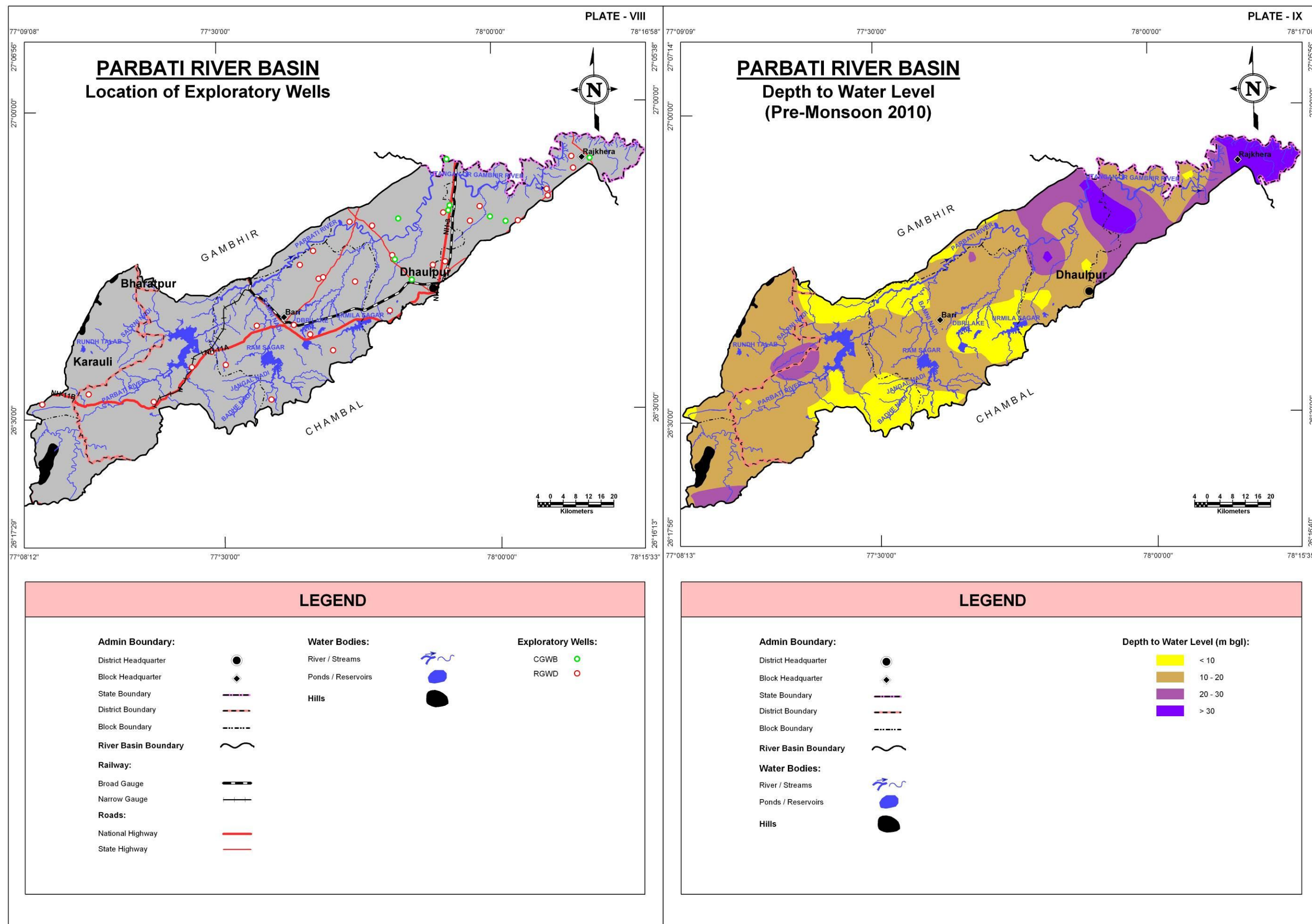
District Name	CGWB	RGWD	Total
Bharatpur	-	-	-
Dhaulpur	9	31	40
Karauli	-	1	1
Total	9	32	41

DEPTH TO WATER LEVEL (PRE MONSOON – 2010)

The general depth to water level in the basin is less than 20 meters below ground level, as seen in most of the central and western parts of the basin. In the eastern part of the basin, however, there are pockets wherein deeper water levels were encountered. The two areas, first one in the north and northeast of Dhaulpur city and the second one around Rajkhera, the depth to water level is highest i.e., reaching upto 30m bgl or more.

Depth to water level (m bgl) Pre Monsoon - 2010	District wise area coverage (sq km)*			Total Area (sq km)
	Bharatpur	Dhaulpur	Karauli	
< 10	4.7	398.6	4.2	407.5
10 - 20	2.4	1,063.8	319.8	1,386.0
20 - 30	-	283.4	44.4	327.8
> 30	-	128.0	-	128.0
Total	7.1	1,873.8	368.4	2,249.3

* The area covered in the derived maps is less than the total basin area since the hills have been excluded from interpolation/contouring.



WATER TABLE ELEVATION (PRE MONSOON 2010)

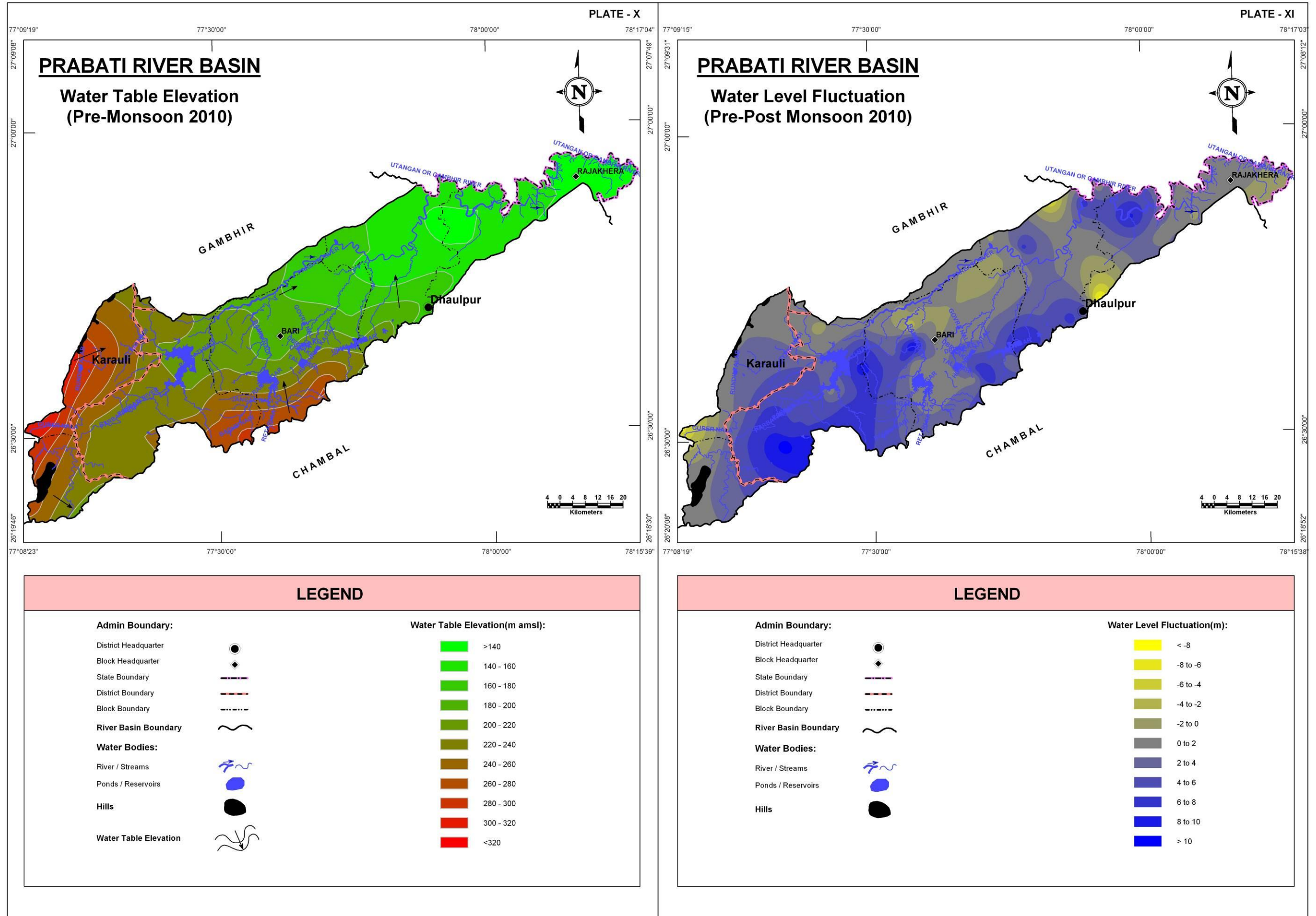
The flow directions marked on water table elevation contour map (Plate – X) indicate an initial west and southeast to northwards and northeastwards general flow of ground water within the Parbati river basin. The highest Water table elevation is seen in the western hilly area in Karauli district attaining >320m amsl, whereas the lowest elevation (about 140m amsl) is recorded at the end of river basin where the river leaves the basin in Dhaulpur district near Rajakhera. That shows a variation of more than 180m of water table elevation in the basin. The flow gradient is also steep in the hilly areas but becomes sluggish in plain areas of Dhaulpur district.

Water Table Elevation (m amsl) (Pre Monsoon – 2010)	District wise area coverage (sq km)			Total Area (sq km)
	Banswara	Chittaurgarh	Dungarpur	
>140	-	151.5	-	151.5
140 - 160	-	387.9	-	387.9
160 - 180	-	241.9	-	241.9
180 - 200	-	215.4	-	215.4
200 - 220	-	251.7	6.6	258.3
220 - 240	7.1	383.0	62.2	452.3
240 - 260	-	142.3	127.2	269.5
260 - 280	-	90.8	90.0	180.8
280 - 300	-	9.3	61.7	71.0
300 - 320	-	-	20.5	20.5
< 320	-	-	0.2	0.2
Total	7.1	1,873.8	368.4	2,249.3

WATER LEVEL FLUCTUATION (PRE TO POST MONSOON 2010)

Water level fluctuation map is presented in Plate – XI. The fluctuation range is high as seen from the values of about -8m to more than 10m which is about 18m. The predominantly blue colour in the map suggests that the water levels in the post monsoon season have shown, in general, rise with respect to pre-monsoon water levels in the year 2010. Barring the steep rise (as seen in the southern part of Dhaulpur district) or fall (north of Dhaulpur city) in water levels which however occupy limited areas in the basin, the fluctuation has largely remained in the range of -2m to 8m.

District Name	District wise area coverage (sq km) within fluctuation range (m)											Total Area (sq km)
	< -8	-8 to -6	-6 to -4	-4 to -2	-2 to 0	0 to 2	2 to 4	4 to 6	6 to 8	8 to 10	> 10	
Bharatpur	-	-	-	-	-	7.1	-	-	-	-	-	7.1
Dhaulpur	0.2	1.9	5.8	17.1	167.7	613.6	404.6	384.8	221.6	52.0	4.5	1,873.8
Karauli	-	0.6	4.4	9.4	22.0	142.2	150.3	38.2	1.3	-	-	368.4
Total	0.2	2.5	10.2	26.5	189.7	762.9	554.9	423.0	222.9	52.0	4.5	2,249.3



ELECTRICAL CONDUCTIVITY DISTRIBUTION

PARBATI RIVER BASIN

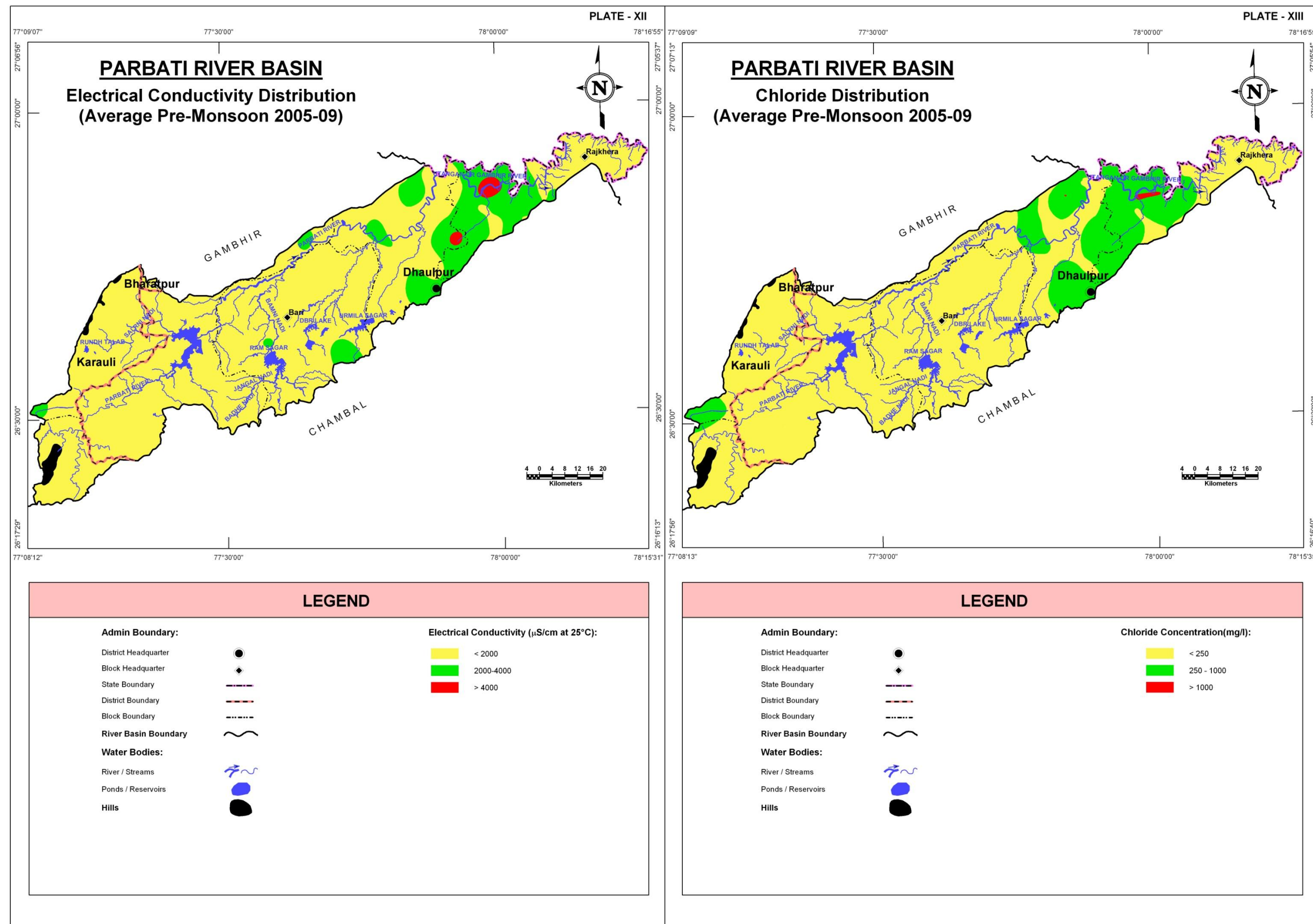
The Electrical Conductivity (at 25°C) distribution map is presented in Plate – XII reveals an in general low values in the basin which makes it suitable for domestic purposes. The only strip between Dhaulpur and Rajkhera in the northern part of the basin has shown the presence of moderate to high EC (>2000 $\mu\text{S}/\text{cm}$) in ground water which limits to some extent, its suitability for different purposes. Hardly 0.02% of the basin area falls within high EC (>4000 $\mu\text{S}/\text{cm}$) region.

Electrical Conductivity Ranges (μS/cm at 25°C) (Ave. of years 2005-09)	District wise area coverage (sq km)						Total Area (sq km)
	Bharatpur		Dhaulpur		Karauli		
	Area	% age	Area	% age	Area	% age	
< 2000	7.08	100.0	1576	84.1	362.6	98.4	1,945.7
2000-4000	-	-	281.1	15.0	5.8	1.6	286.9
> 4000	-	-	16.7	0.9	-	-	16.7
Total	7.1	100.0	1,873.8	100.0	368.4	100.0	2,249.3

CHLORIDE DISTRIBUTION

High chloride concentration in ground water also renders it unsuitable for domestic and other purposes. The chloride concentration map (Plate – XIII) exhibits a similar distribution pattern as to that of Electrical Conductivity. A predominantly yellow colour (<250 mg/l) in the map suggests that from chloride perspective, the ground water is suitable for all purposes and together with moderately high concentration (250 – 1000 mg/l), these areas occupy more than 99.3% of basin area and the rest of the area (0.7%) in the north has shown high chloride concentration in ground water.

Chloride Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)						Total Area (sq km)
	Bharatpur		Dhaulpur		Karauli		
	Area	% age	Area	% age	Area	% age	
< 250	7.08	100.0	1576	84.1	362.6	98.4	1,945.7
250 - 1000	-	-	281.1	15.0	5.8	1.6	286.9
> 1000	-	-	16.7	0.9	-	-	16.7
Total	7.1	100.0	1,873.8	100.0	368.4	100.0	2,249.3



FLUORIDE DISTRIBUTION

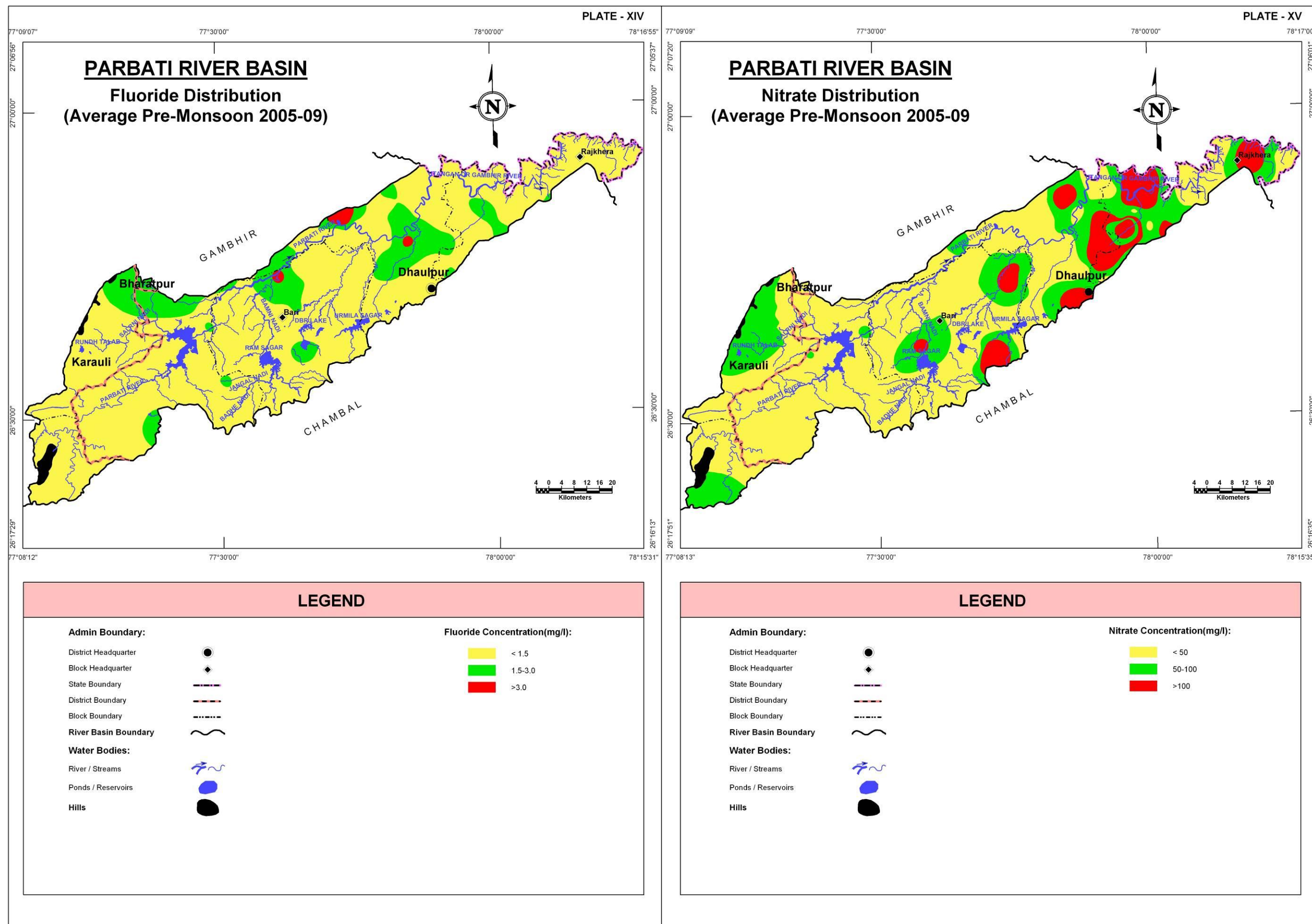
The Fluoride concentration map for Parbati River Basin (Plate – XIV) shows three small patches in the central part of the basin where very high Fluoride concentration (>3 mg/l) was recorded and these occupy an insignificant (less than 1%) part of basin area. Moderate Fluoride concentration (1.5 – 3.0 mg/l) occupies about 15% of basin area surrounding the high concentration areas. The rest of the area (approximately 85%) has shown very low concentration in ground water samples which makes it suitable for all purposes.

Fluoride Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)						Total Area (sq km)
	Bharatpur		Dhaulpur		Karauli		
	Area	% age	Area	% age	Area	% age	
< 1.5	-	-	1,570.0	83.8	330.9	89.8	1,900.9
1.5-3.0	7.1	100.0	287.1	15.3	37.5	10.2	331.7
> 3.0	-	-	16.7	0.9	-	-	16.7
Total	7.1	100.0	1,873.8	100.0	368.4	100.0	2,249.3

NITRATE DISTRIBUTION

High nitrate concentration in ground water renders it unsuitable for agriculture purposes. Plate – XV shows distribution of Nitrate in ground water. The distribution pattern has a different trend from EC distribution and several isolated high Nitrate concentration (>100 mg/l) patches are seen distributed over the area located in the northern and central parts of the basin. Such areas of high Nitrate concentration in ground water constitute approximately 10% of the basin area only. Moderate concentration of Nitrate is found in approximately 24% of the basin area. Rest of the basin in southern and central parts, has low Nitrate concentration in ground water and thus rendering it suitable for agriculture.

Nitrate Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)						Total Area (sq km)
	Bharatpur		Dhaulpur		Karauli		
	Area	% age	Area	% age	Area	% age	
< 50	7.1	100.0	1,279.6	68.3	223.1	60.6	1,509.8
50-100	-	-	384.4	20.5	145.3	39.4	529.7
> 100	-	-	209.8	11.2	-	-	209.8
Total	7.1	100.0	1,873.8	100.0	368.4	100.0	2,249.3



DEPTH TO BEDROCK

Below soil cover and thick pile alluvial material at different depths, and weathered/fractured bedrock material which constitutes good aquifer material, the massive bedrock is found that is not very significant from ground water occurrence perspective. Plate – XVI represents depth to bedrock in meters below ground level. The bedrock is generally found at moderate depths ranging from about 20m bgl to about 80m bgl as seen by regions marked on map which together constitute about 95% of the basin area. The rest of the basin (approximately 5%) is either shallower than 20m or deeper than 80m.

Depth to Bedrock (m bgl)	District wise area coverage (sq km)			Total Area (sq km)
	Bharatpur	Dhaulpur	Karauli	
< 20	-	33.6	-	33.6
20-40	-	268.7	34.3	303.0
40-60	6.8	871.0	211.1	1,088.9
60-80	0.3	666.0	99.3	765.6
> 80	-	34.5	23.7	58.2
Total	7.1	1,873.8	368.4	2,249.3

UNCONFINED AQUIFER

Hydrogeological properties are different for alluvial and hard rock aquifers and therefore, this aquifer has been mapped as two separate regions viz, unconfined aquifers in alluvial and in hard rock areas.

The alluvial material is predominantly alluvial or fluvial origin sand, clay and gravel. The thickness of this aquifer varies from very thin layers to upto 30m in the northern parts of Dhaulpur district. The hardrock aquifers are formed in weathered and fractured part of the hardrock that can sustain water under unconfined conditions. Such aquifers are formed in Vindhyan sandstone rocks in southern parts of Dhaulpur district and in Karauli district. These aquifers significantly vary in thickness ranging from less than 10m to about 90m.

Alluvial areas

Unconfined aquifer Thickness (m)	District wise area coverage (sq km)			Total Area (sq km)
	Bharatpur	Dhaulpur	Karauli	
<10	-	595.5	-	595.5
10-20	-	229.5	-	229.5
>20	-	4.2	-	4.2
Total	-	829.2	-	829.2

Hardrock areas:

Unconfined aquifer Thickness (m)	District wise area coverage (sq km)			Total Area (sq km)
	Bharatpur	Dhaulpur	Karauli	
< 10	-	129.2	-	129.2
10-20	7.1	148.1	10.2	165.4
20-30	-	173.2	57.2	230.4
30-40	-	220.3	83.5	303.8
40-50	-	152.0	79.7	231.7
50-60	-	95.2	58.2	153.4
60-70	-	60.5	41.0	101.5
70-80	-	38.0	26.8	64.8
80-90	-	21.0	11.5	32.5
> 90	-	7.1	0.3	7.4
Total	7.1	1,044.6	368.4	1,420.1

CROSS SECTIONS

PARBATI RIVER BASIN

Several hydrogeologic cross sections have been drawn to better depict the sub-surface distribution of lithology. These sections have been overlaid with geological maps and structural faults if there are any have been transferred for verification of their impact on sub-surface material disposition. The alignment of the cross sections is shown in Plate – XVIII and corresponding sections are presented in Plates – XIX to XXI. The broad alignment of the sections is as given below:

Name of Section Line	Orientation
Section AA'	SW – NE
Section BB'	SW – NE
Section CC'	W – E

CROSS SECTIONS

PARBATI RIVER BASIN

Section A-A':

The section AA' (Plate – XIX) is taken along SW-NE direction and is prepared based on the lithologs of 9 boreholes, which cover length of about 80 Km. In the southwestern part, sandstone formation underlying top soil occurs for about a length of 40Km along the profile and gradually goes deeper giving way to progressively increasing thicker pile of alluvium. While sand is predominant material in the alluvium, there are frequent intercalations with clay formation which gradually increase in thickness northeastwards.

Water table varies from 140 m amsl to 215 m amsl in the section.

Section B-B':

The section BB' (Plate – XX) is also drawn in the SW-NE direction based on the lithologs of 5 boreholes and covering a distance of about 40Km. In the southwest, the section is dominated by 30-40m of sandy alluvial material underlain by sandstone. The sandstone terminates in the middle of the section till a length of 22 km and giving way to still thicker pile of alluvial material that is sandy in nature. In the alluvial cover, the sand is interbedded with clay and also there is a consistent soil cover above all these.

Water table varies in elevation from 140 m amsl to 165 m amsl in the section.

PLATE - XIX

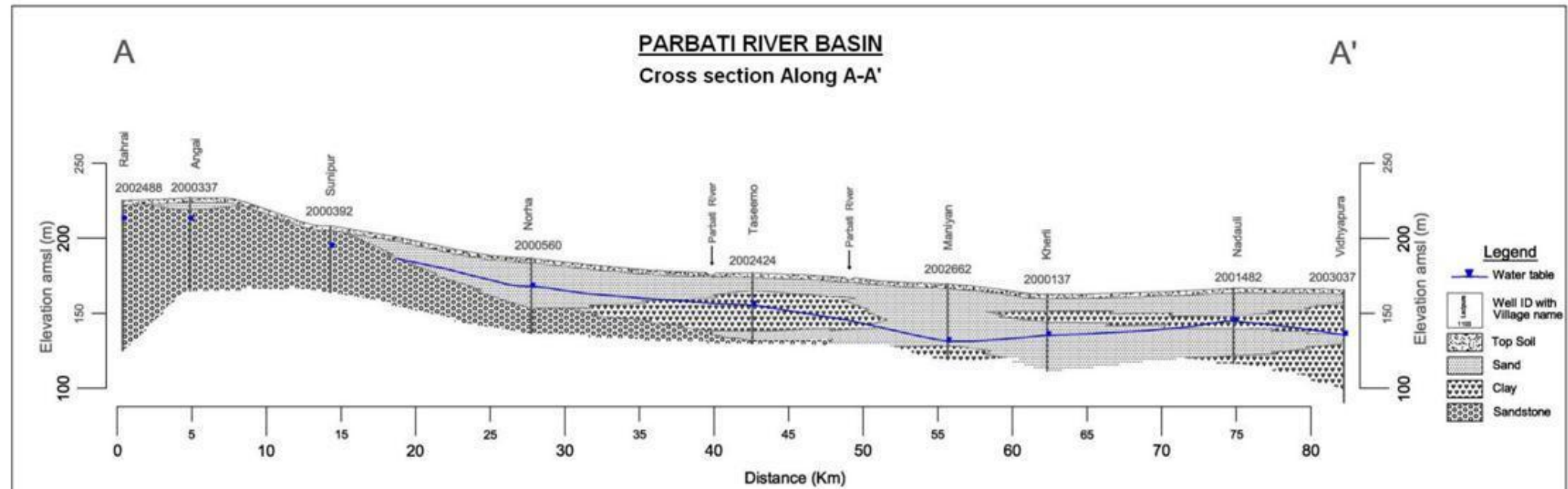
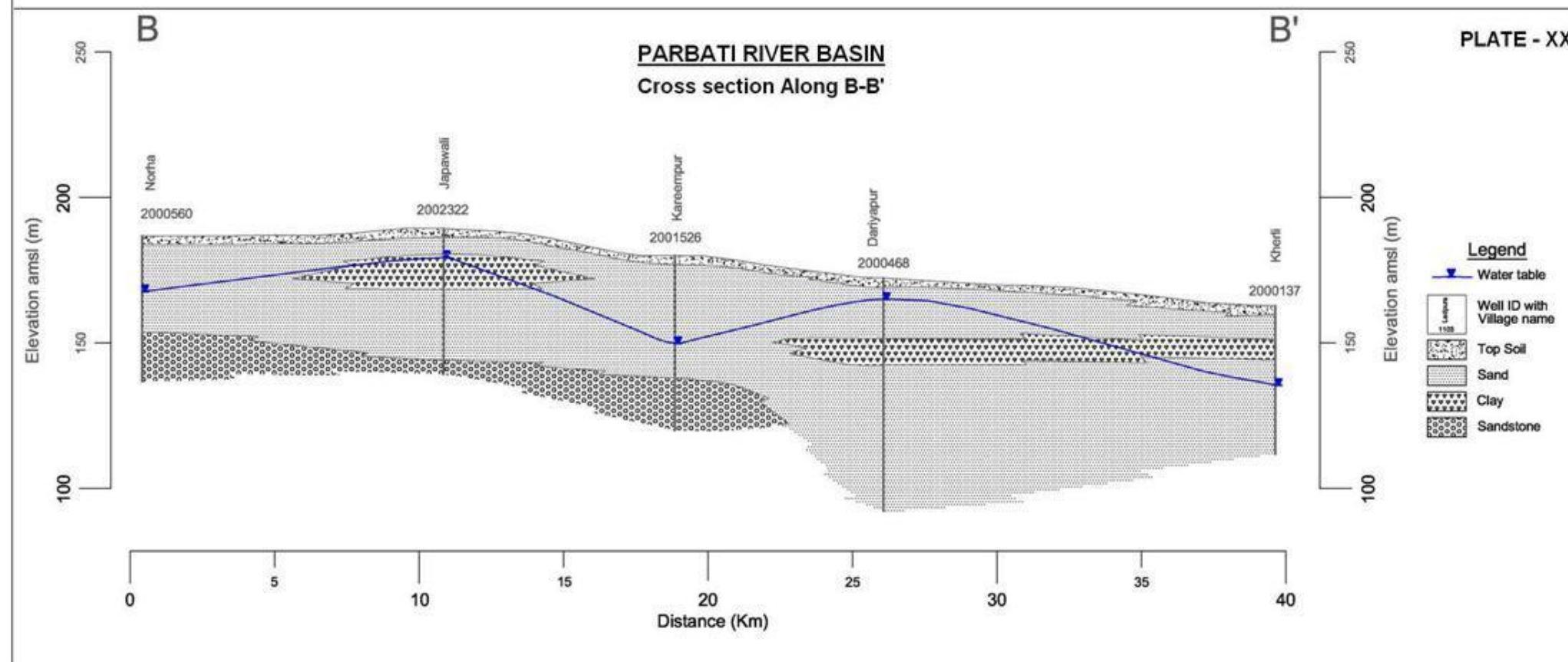


PLATE - XX



CROSS SECTIONS

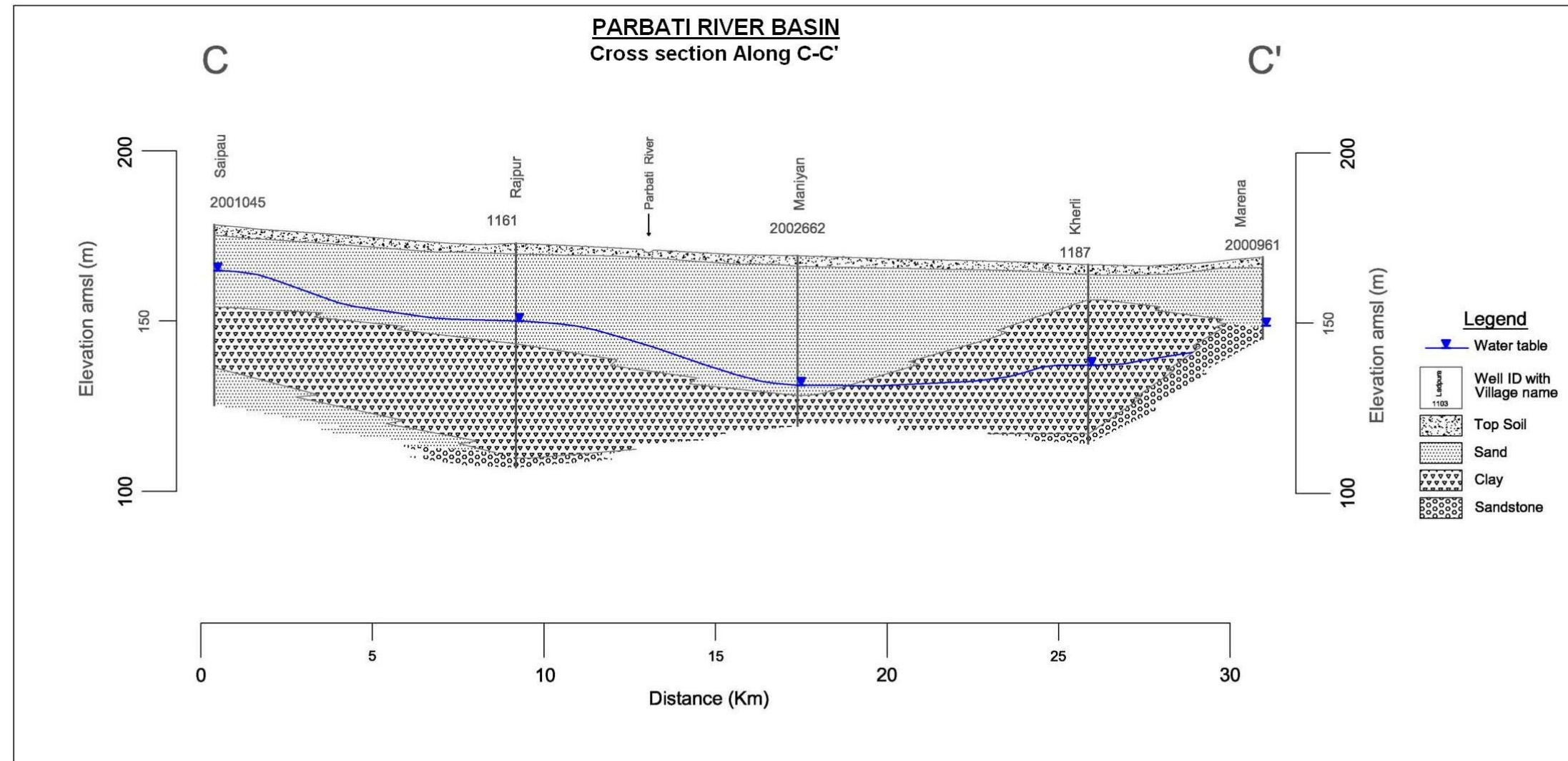
PARBATI RIVER BASIN

Section C-C':

The Section CC' (Plate – XXI) has been chosen in W-E direction, in the northern part of the river basin, broadly parallel to the Parbati River. This section is prepared based on the lithologs of 5 boreholes and extends over a length of 30Km. The section reveals the presence of sandstone forming the bedrock at shallower depths in the east which is deeper further west. The alluvium also increases in thickness westwards reaching to more than 50m in thickness. Thick layer of clay rests over sandstone and in turn overlain by variable but persistent sand below a thin layer of soil cover. Sandstone was encountered in the well of Kheril, Marena and Rajpur.

Water table is variable in elevation ranging from 150 m amsl to 170 m amsl in the section.

PLATE - XXI



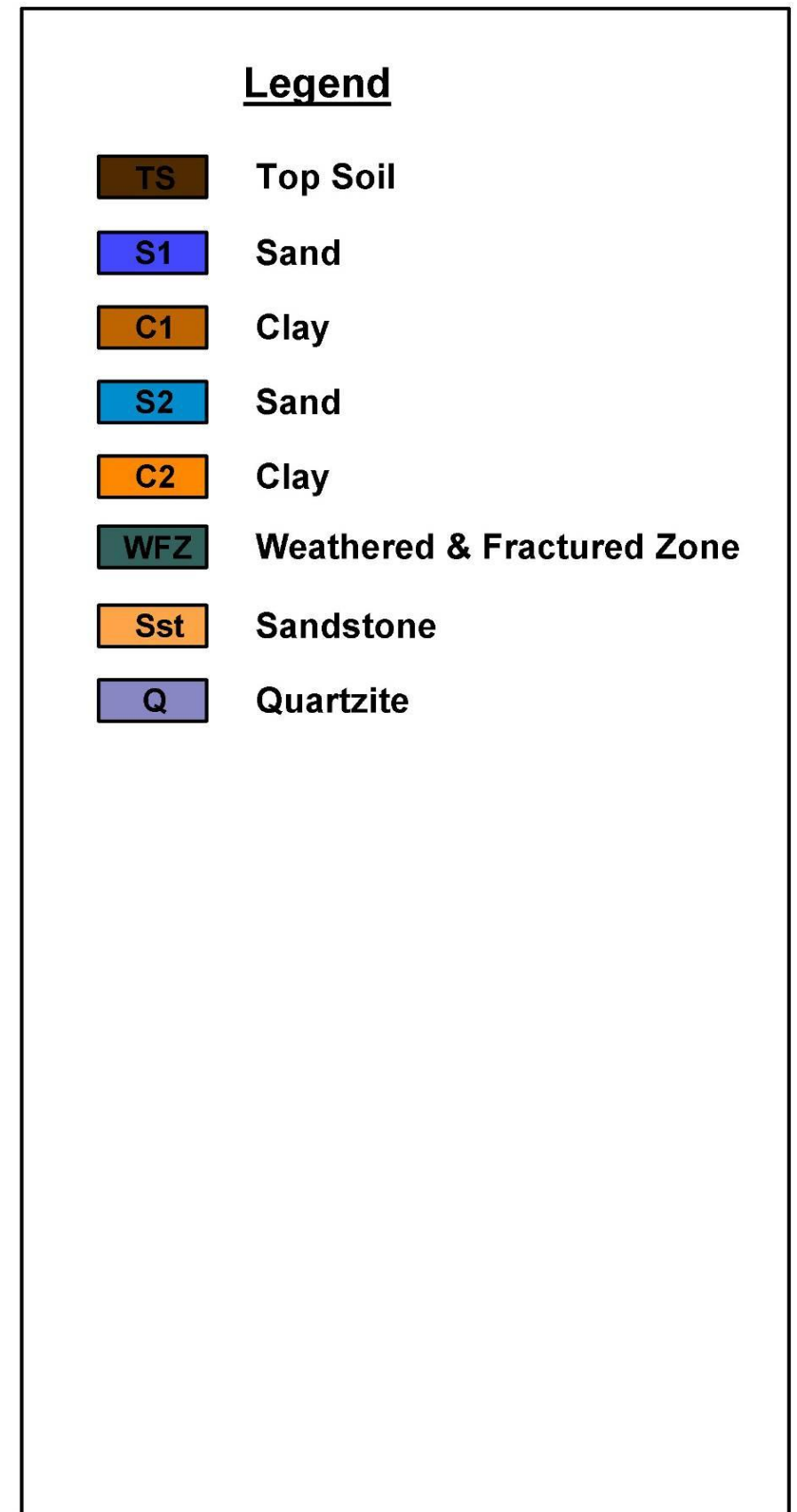
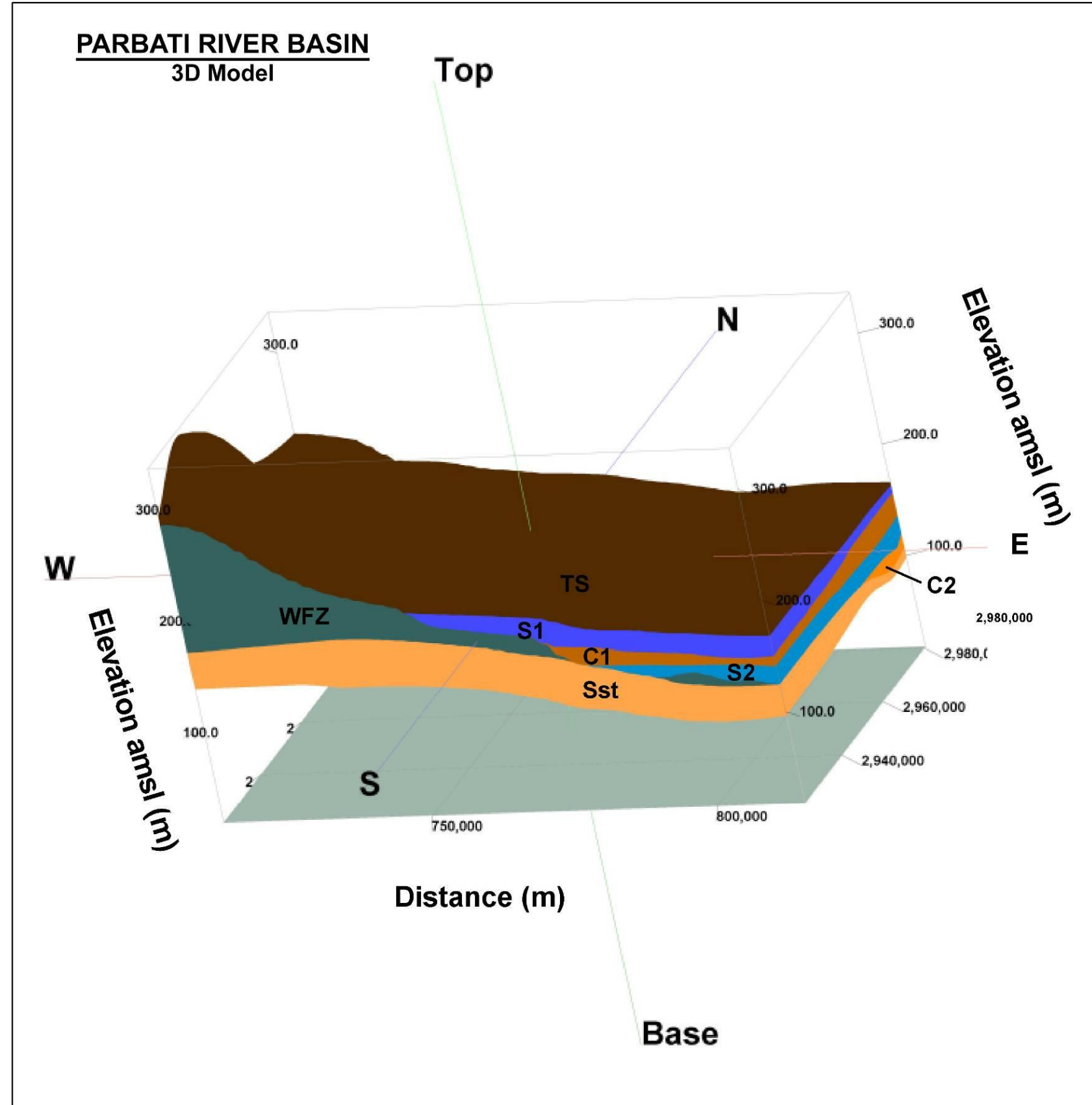
3D MODEL OF AQUIFERS

PARBATI RIVER BASIN

The continuous litho-stratigraphic model has been developed for the Parbati River Basin using the data of scattered wells as input. 3D model depicts the sub-surface aquifer disposition of litho-stratigraphic units forming aquifers, aquicludes and aquitards in the area.

Plate XXII presents 3D model depicting the various litho-stratigraphic units in the entire river basin. From this model it appears that beneath the soil cover, there are two persistent clay horizons in the region separating two sandy aquifers. The second sandy aquifer is found only in the eastern part of the basin within alluvium. The depth of bedrock is less in the eastern part as compared to the western part of the basin. Sandstone appears in entire basin overlain by weathered & fractured rock in western part and second sandy aquifer in eastern part of the basin.

PLATE - XXII

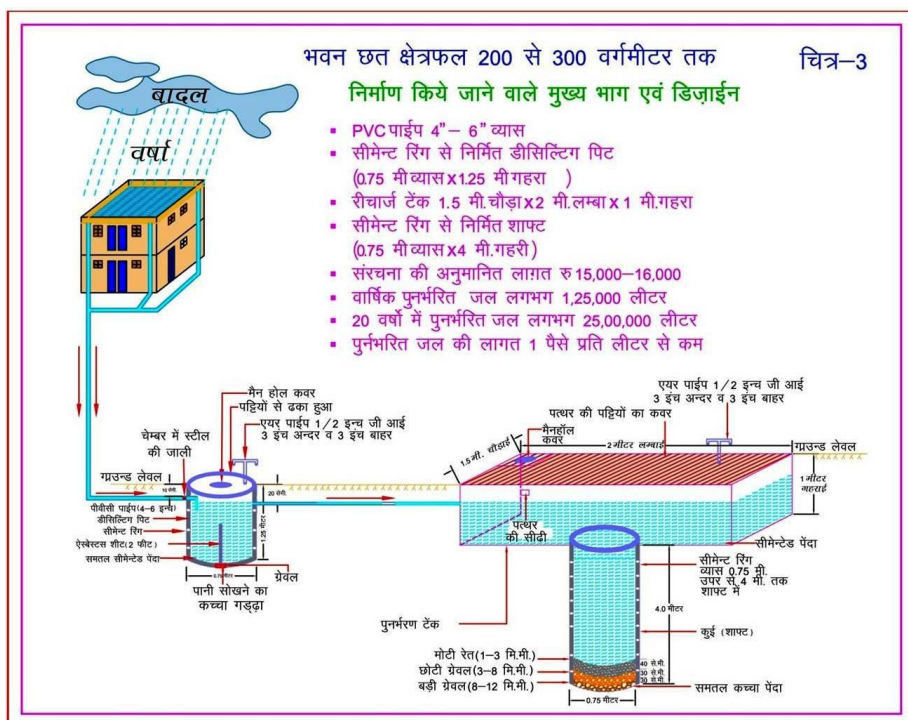
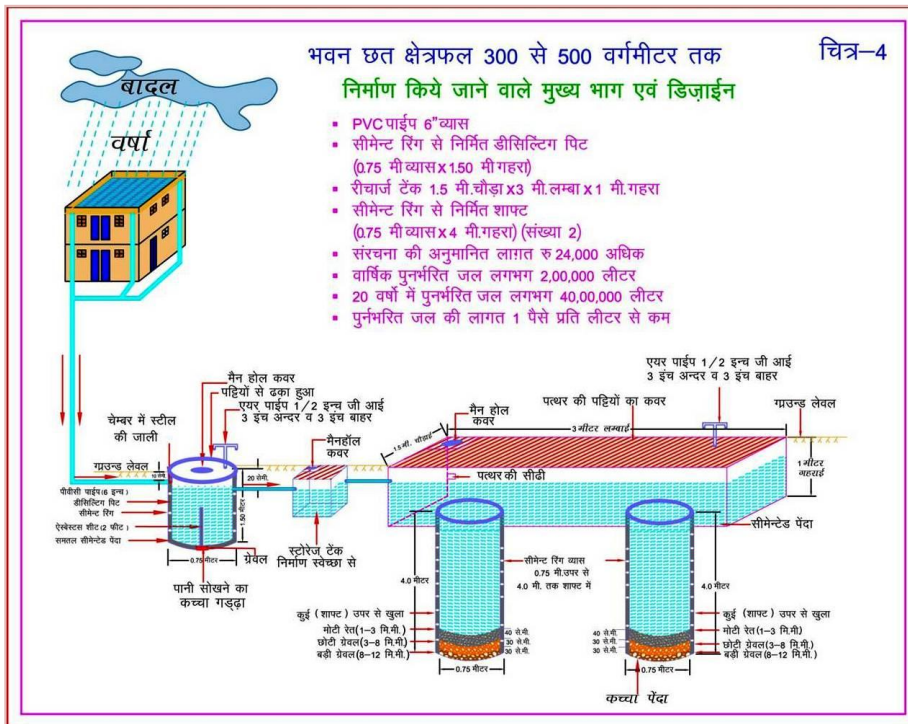


Glossary of terms

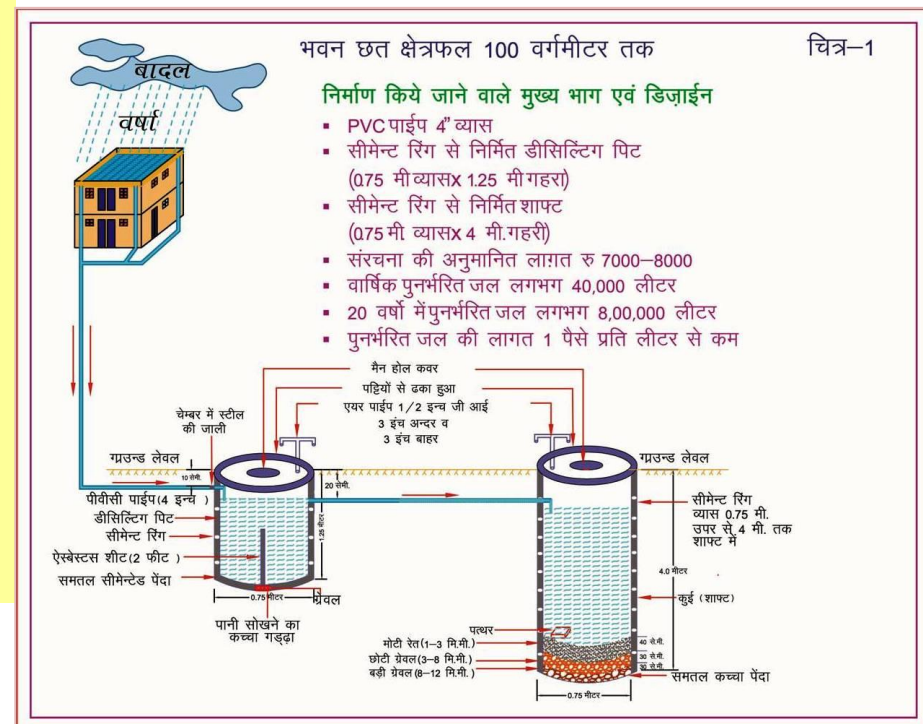
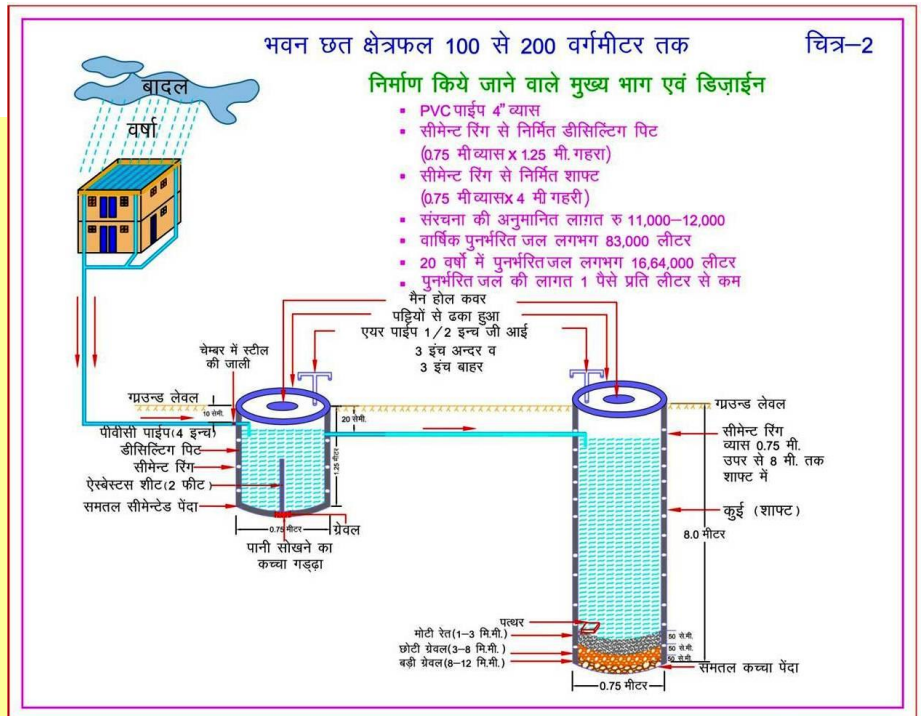
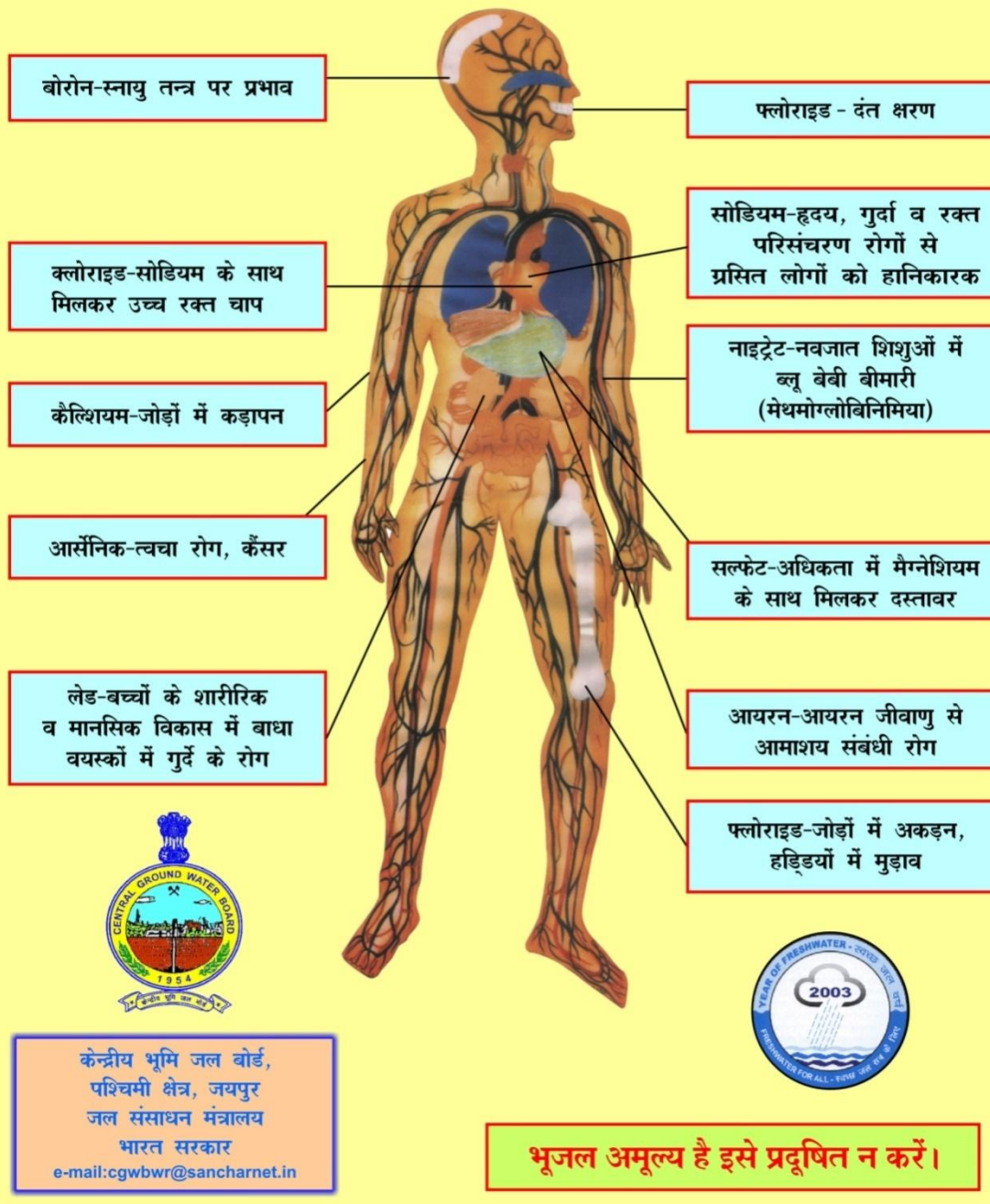
S. No.	Technical Terms	Definition
1	AQUIFER	A saturated geological formation which has good permeability to supply sufficient quantity of water to a Tube well, well or spring.
2	ARID CLIMATE	Climate characterized by high evaporation and low precipitation.
3	ARTIFICIAL RECHARGE	Addition of water to a ground water reservoir by man-made activity
4	CLIMATE	The sum total of all atmospheric or meteorological influences principally temperature, moisture, wind, pressure and evaporation of a region.
5	CONFINED AQUIFER	A water bearing strata having confined impermeable overburden. In this aquifer, water level represents the piezometric head.
6	CONTAMINATION	Introduction of undesirable substance, normally not found in water, which renders the water unfit for its intended use.
7	DRAWDOWN	The drawdown is the depth by which water level is lowered.
8	FRESH WATER	Water suitable for drinking purpose.
9	GROUND WATER	Water found below the land surface.
10	GROUND WATER BASIN	A hydro-geologic unit containing one large aquifer or several connected and interrelated aquifers.
11	GROUND WATER RECHARGE	The natural infiltration of surface water into the ground.
12	HARD WATER	The water which does not produce sufficient foam with soap.
13	HYDRAULIC CONDUCTIVITY	A constant that serves as a measure of permeability of porous medium.
14	HYDROGEOLOGY	The science related with the ground water.
15	HUMID CLIMATE	The area having high moisture content.
16	ISOHYET	A line of equal amount of rainfall.
17	METEOROLOGY	Science of the atmosphere.
18	PERCOLATION	It is flow through a porous substance.
19	PERMEABILITY	The property or capacity of a soil or rock for transmitting water.
20	pH	Value of hydrogen-ion concentration in water. Used as an indicator of acidity (pH < 7) or alkalinity (pH > 7).
21	PIEZOMETRIC HEAD	Elevation to which water will rise in a piezometers.
22	RECHARGE	It is a natural or artificial process by which water is added from outside to the aquifer.
23	SAFE YIELD	Amount of water which can be extracted from ground water without producing undesirable effect.
24	SALINITY	Concentration of dissolved salts.
25	SEMI-ARID	An area is considered semiarid having annual rainfall between 10-20 inches.
26	SEMI-CONFINED AQUIFER	Aquifer overlain and/or underlain by a relatively thin semi-pervious layer.
27	SPECIFIC YIELD	Quantity of water which is released by a formation after its complete saturation.
28	TOTAL DISSOLVED SOLIDS	Total weight of dissolved mineral constituents in water per unit volume (or weight) of water in the sample.

(Contd...)

S. No.	Technical Terms	Definition
29	TRANSMISSIBILITY	It is defined as the rate of flow through an aquifer of unit width and total saturation depth under unit hydraulic gradient. It is equal to product of full saturation depth of aquifer and its coefficient of permeability.
30	UNCONFINED AQUIFER	A water bearing formation having permeable overburden. The water table forms the upper boundary of the aquifer.
31	UNSATURATED ZONE	The zone below the land surface in which pore space contains both water and air.
32	WATER CONSERVATION	Optimal use and proper storage of water.
33	WATER RESOURCES	Availability of surface and ground water.
34	WATER RESOURCES MANAGEMENT	Planned development, distribution and use of water resources.
35	WATER TABLE	Water table is the upper surface of the zone of saturation at atmospheric pressure.
36	ZONE OF SATURATION	The ground in which all pores are completely filled with water.
37	ELECTRICAL CONDUCTIVITY	Flow of free ions in the water at 25C mu/cm.
38	CROSS SECTION	A Vertical Projection showing sub-surface formations encountered in a specific plane.
39	3-D PICTURE	A structure showing all three dimensions i.e. length, width and depth.
40	GWD	Ground Water Department
41	CGWB	Central Ground Water Board
42	CGWA	Central Ground Water Authority
43	SWRPD	State Water Resources Planning Department
44	EU-SPP	European Union State Partnership Programme
45	TOPOGRAPHY	Details of drainage lines and physical features of land surface on a map.
46	GEOLOGY	The science related with the Earth.
47	GEOMORPHOLOGY	The description and interpretation of land forms.
48	PRE MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer before Monsoon (carried out between 15th May to 15th June)
49	POST-MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer after Monsoon (carried out between 15th October to 15th November)
50	PIEZOMETER	A non-pumping small diameter bore hole used for monitoring of static water level.
51	GROUND WATER FLUCTUATION	Change in static water level below ground level.
52	WATER TABLE	The static water level found in unconfined aquifer.
53	DEPTH OF BED ROCK	Hard & compact rock encountered below land Surface.
54	G.W. MONITORING STATION	Dug wells selected on grid basis for monitoring of state water level.
55	EOLIAN DEPOSITS	Wind-blown sand deposits



भूजल में घुले मुख्य तत्वों की अधिकता का मानव शरीर पर दुष्प्रभाव



Myths and Facts about Ground Water

S No	Myths	Facts
1	What is Ground Water • an underground lake • a net work of underground rivers • a bowl filled with water	Water which occurs below the land in geological formations/rocks is Ground water
2	Ground Water occurs everywhere beneath the Land Surface	Not really, it depends on the nature of rock formation
3	There is a relationship between ground water and surface water	Not all the places. Near streams/rivers there is relation
4	Groundwater is not renewable resource	It is renewable source and every year it is being recharged through rain/applied irrigation etc
5	Ground water is unlimited and deeper you drill more discharge	It is limited to annual recharge from rain/applied irrigation. The discharge may not increase if you go deeper
6	Ground Water moves rapidly	The movement of ground water is very slow
7	Ground water pumped from wells is thousands of years old	Generally the ground water being tapped through wells is a few years old
8	If water taste good—it is safe to drink	It may have other chemicals e.g. fluoride, nitrates etc which are harmful
9	Water from free flowing tube wells is very pure	This water can also be contaminated so test before use
10	If I recharge my TW/DW/HP it will not benefit me	It will also benefit you and also adjoining wells
11	There is no static ground water resources in Rajasthan	Rajasthan is also having Static GW resources, and being tapped in most of areas as GW annual withdrawal is more than annual recharge
12	I cannot meet annual cooking and drinking water requirement by rain water harvesting	The water requirement for drinking and cooking is only 8 lit/day. You can harvest this water for family of 5 persons from roof top or paved area of 75 Sq m to meet annual requirement
13	You can increase ground water recharge	This can be done by harvesting the rain water and storing in sub surface reservoir (GW) by constructing the recharge structures
14	You cannot use abandoned TW/HP/DW for ground water recharge	These should be used as recharge structures as harvested rain water is directly put into GW reservoir
15	Putting waste near HP/TW will not cause any problem	Such actions will pollute wells and water



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